

When Does Density Beat Centroids? A Comparative Study of K-Means and DBSCAN-Family Clustering for Spatial Hotspot Detection

Data Mining Course Project

Abstract

We compare four clustering algorithms — K-Means with random initialisation, K-Means++, DBSCAN and HDBSCAN — on three datasets drawn from two application settings: spatial hotspot detection, where no ground truth exists, and protein localisation, where it does. Our motivating application is the discovery of spatially coherent activity zones from point events, evaluated on 23,412 global earthquake epicentres and 6,000 New York City ride-hailing pickups. A third dataset, the 336-protein *Ecoli* localisation benchmark, supplies the ground-truth labels the spatial data lacks. Three findings emerge. First, the algorithms disagree not marginally but categorically: on the earthquake data K-Means partitions the globe into three convex wedges that cut across tectonic boundaries, while DBSCAN recovers 68 arc-shaped boundary segments while discarding only 2% of events as noise. Second, and more consequentially, on the labelled dataset the silhouette coefficient ranks the four methods in almost exactly the reverse order of the Adjusted Rand Index (rank correlation -0.80): the method with the best silhouette, HDBSCAN, has the worst ARI, and the method with the worst silhouette, K-Means++, has the best ARI. Since internal indices are the only selection tool available on unlabelled spatial data, this result directly undermines the standard practice of choosing clustering parameters by maximising silhouette. Third, a dimensionality ablation reveals a clean crossover: DBSCAN outperforms K-Means++ at two and three dimensions (ARI 0.575 and 0.618 versus 0.468 and 0.466) and loses decisively at seven (0.455 versus 0.731). We conclude that algorithm choice is governed by cluster geometry and intrinsic dimensionality rather than by any internal quality score, and we give an explicit selection protocol that makes the operational requirements of the application, not the metric, the binding constraint.

1 Introduction

1.1 Motivation

Many operational decisions reduce to the same question: given a cloud of recorded events, where are the concentrations, how many are there, and which events belong to no concentration at all? A seismologist asks it of earthquake epicentres in order to delineate active fault systems. A fleet operator asks it of ride pickups in order to decide where to position idle vehicles. In neither case is the number of groups known in advance, in neither case are the groups convex, and in both cases a substantial fraction of events are genuinely isolated and should not be forced into any group.

These three properties — unknown cluster count, arbitrary shape, and the existence of true noise — are precisely the assumptions that separate centroid-based clustering from density-based clustering. K-Means requires the analyst to supply k , implicitly models every cluster as an isotropic ball, and assigns every point to some cluster. DBSCAN requires neither k nor convexity and labels low-density points as noise. The application therefore does not merely provide a testbed for comparing the two families; it predicts which should win. Our experiments test that prediction.

1.2 Problem statement

A comparison of this kind faces an awkward obstacle. The datasets where the question is operationally interesting are exactly the datasets where no ground truth exists — nobody has labelled

which earthquake belongs to which fault segment. Evaluation must therefore fall back on internal validity indices such as the silhouette coefficient. But the silhouette is defined through within-cluster and between-cluster distances to *all* other points of a cluster, a formulation that rewards compact, well-separated, roughly spherical groups. It therefore encodes the assumption made by one of the two algorithm families under comparison. Using it as the arbiter is not a neutral act.

This paper takes that tension as its subject. We ask not only which algorithm recovers better structure, but whether the metrics available in the unlabelled setting would have told us so.

1.3 Contributions

- A unified comparison of four clustering algorithms across three datasets spanning two application domains, with a single evaluation harness and an explicit, application-driven parameter-selection protocol (Section 4).
- Quantification of a systematic disagreement between internal and external validity indices on labelled data (rank correlation -0.80), and a demonstration that the same mechanism corrupts parameter selection on unlabelled spatial data (Sections 5.4 and 5.3).
- Four ablations isolating initialisation, distance metric, feature scaling and intrinsic dimensionality, including a crossover point at which the ranking of the two algorithm families reverses (Section 6).
- A practical selection rule derived from the failure modes observed, rather than from metric values (Section 7.4).

2 Related Work

Centroid-based clustering. Lloyd’s algorithm for k -means minimises within-cluster sum of squares by alternating assignment and centroid update; it converges to a local optimum whose quality depends heavily on initialisation. Arthur and Vassilvitskii’s k -means++ seeding selects initial centres with probability proportional to squared distance from the nearest existing centre, achieving an $O(\log k)$ expected approximation to the optimal objective. We treat the two as an initialisation ablation rather than as separate algorithms, since they share an identical update rule.

Density-based clustering. DBSCAN (Ester et al., 1996) defines clusters as maximal sets of density-connected points, where a point is a core point if at least `minPts` points lie within ε . Points reachable from no core point are labelled noise. The method needs no k and imposes no shape constraint, but the single global ε assumes uniform density across the dataset. OPTICS relaxes this by producing a reachability ordering rather than a flat partition; HDBSCAN (Campello et al., 2013) goes further, building a hierarchy over the mutual-reachability distance and extracting a flat clustering by maximising cluster stability, which removes ε entirely and leaves only a minimum cluster size.

Validity indices. Internal indices — silhouette, Davies–Bouldin, Calinski–Harabasz — score a partition using only the data and the labels. External indices — Adjusted Rand Index, Normalised Mutual Information — compare against ground truth. It is known in principle that internal indices embed geometric assumptions; our contribution is to measure how large the resulting distortion is in a setting where only internal indices are available.

3 Methods

3.1 K-Means and K-Means++

K-Means seeks centroids $\{\mu_1, \dots, \mu_k\}$ minimising $J = \sum_{i=1}^n \min_j \|x_i - \mu_j\|^2$. Each iteration assigns every point to its nearest centroid and recomputes centroids as cluster means, giving $O(nkdi)$ time

Table 1: Structural assumptions of the four methods.

	K-Means	K-Means++	DBSCAN	HDBSCAN
Cluster shape assumed	isotropic	isotropic	arbitrary	arbitrary
Requires k	yes	yes	no	no
Explicit noise label	no	no	yes	yes
Handles varying density	no	no	no	yes
Parameter is interpretable	no	no	yes (ε in km)	partly
Deterministic	no	no	yes	yes
Time complexity	$O(nkdi)$	$O(nkdi)$	$O(n \log n)$	$O(n \log n)$

for i iterations. The objective is exactly the sum of squared Euclidean distances, so the implied cluster model is an isotropic Gaussian of equal variance; non-convex or elongated structure cannot be represented. Every point receives a label, so outliers displace centroids rather than being isolated. K-Means++ modifies only the initial centre selection.

3.2 DBSCAN

DBSCAN takes ε and `minPts`. A point is a *core point* if $|N_\varepsilon(p)| \geq \text{minPts}$; clusters are the transitive closure of direct density-reachability among core points, extended to border points; all remaining points are noise. Complexity is $O(n \log n)$ with a spatial index. The method makes no shape assumption and requires no k , but a single global ε presumes comparable density everywhere — the assumption that fails when a sparse region and a dense region must both be resolved. Critically for our application, ε carries physical units: on spatial data it is a distance in kilometres, and is therefore directly interpretable, whereas k is not.

3.3 HDBSCAN

HDBSCAN replaces the global ε with the *mutual reachability distance*

$$d_{\text{mreach}}(a, b) = \max\{\text{core}_k(a), \text{core}_k(b), d(a, b)\},$$

builds a minimum spanning tree under this metric, condenses the resulting hierarchy using a minimum cluster size, and selects the set of clusters maximising total stability. The only required parameter is `min_cluster_size`. Because different branches of the hierarchy are cut at different density levels, clusters of differing density can coexist — the precise failure mode of DBSCAN.

4 Experimental Setup

4.1 Datasets

Table 2 summarises the three datasets; Figure 1 shows them. Two are spatial and unlabelled, reflecting the application; the third is non-spatial and labelled, and exists to test whether the conclusions drawn from internal metrics on the first two can be trusted.

D1 — Earthquakes. Global epicentres of magnitude ≥ 5.5 recorded between 1965 and 2016. Features are latitude and longitude. Seismicity concentrates along plate boundaries, which form long, curved, branching arcs of strongly varying width and density, with genuinely isolated intraplate events scattered between them. This is a maximally unfavourable geometry for a centroid-based method.

D2 — UberNYC. A random sample of 6,000 ride-hailing pickups in New York City during April 2014, restricted to a bounding box that removes GPS artefacts. Demand concentrates along

Table 2: Datasets. The first two carry no ground truth, which is precisely the condition the application imposes.

ID	Dataset	n	d	Classes	Application question
D1	Earthquakes	23,412	2	—	Which tectonic boundary segments are active?
D2	UberNYC	6,000	2	—	Where should idle vehicles be positioned?
D3	Ecoli	336	7	8	Do sequence signals predict localisation site?

the Manhattan street grid and at airports, producing clusters that differ in density by more than an order of magnitude between Midtown and the outer boroughs.

D3 — Ecoli. 336 proteins described by seven signal-sequence scores computed from the amino acid sequence, labelled with one of eight subcellular localisation sites. The class distribution is severely imbalanced, with several classes containing fewer than five members — a structure that lets us ask whether a method’s noise label discards signal.

4.2 Preprocessing and coordinate handling

Geographic coordinates in degrees are not a metric space: one degree of longitude spans 111 km at the equator and 71 km at 50° latitude. We therefore project both spatial datasets to local kilometres using an equirectangular projection about the data mean latitude, so that ε and the silhouette are expressed in kilometres and the centroid- and density-based methods are compared on the same distance scale. Section 6.2 ablates this choice against raw-degree Euclidean distance and against exact haversine distance. Ecoli features are standardised to zero mean and unit variance; Section 6.3 ablates this.

4.3 Evaluation metrics

We report **internal** indices — silhouette ($\uparrow$), Davies–Bouldin ($\downarrow$), Calinski–Harabasz ($\uparrow$) — computed with noise points excluded, since a noise point belongs to no cluster and its inclusion would penalise density methods arbitrarily. Where ground truth exists we additionally report **external** indices — ARI and NMI ($\uparrow$) — computed with noise retained as its own label, so that discarding points is penalised. Silhouette on D1 is estimated from a random subsample of 5,000 points for tractability. We also report the number of clusters, the noise fraction, and wall-clock runtime.

4.4 Parameter selection protocol

All four methods are tuned under one protocol, so that no method is advantaged.

With ground truth (D3): the parameter maximising ARI is selected. This is an oracle protocol and reports each method’s best achievable performance.

Without ground truth (D1, D2): only internal indices are available. An unconstrained silhouette search, however, collapses to a degenerate solution — on both spatial datasets the silhouette is maximised at $k = 2$, which answers no operational question. We therefore impose the requirements the application itself dictates: a solution must return between 3 and 150 clusters and may discard at most 40% of events as noise. Within that feasible region we maximise silhouette. This constraint is applied identically to K-Means, DBSCAN and HDBSCAN.

ε candidates are generated from the k-distance curve: the distance to the `minPts`-th nearest neighbour is computed for every point, sorted, and the knee located as the point of maximum deviation from the chord joining the curve’s endpoints (Figure 4). The grid spans $0.10\times$ to $3.0\times$ the knee value, crossed with `minPts` $\in \{4, 8, 16, 32\}$ (D1, D2) or $\{3, 5, 8, 12\}$ (D3). For K-Means, k ranges over $2 \dots 25$; for HDBSCAN, `min_cluster_size` over a geometric grid.

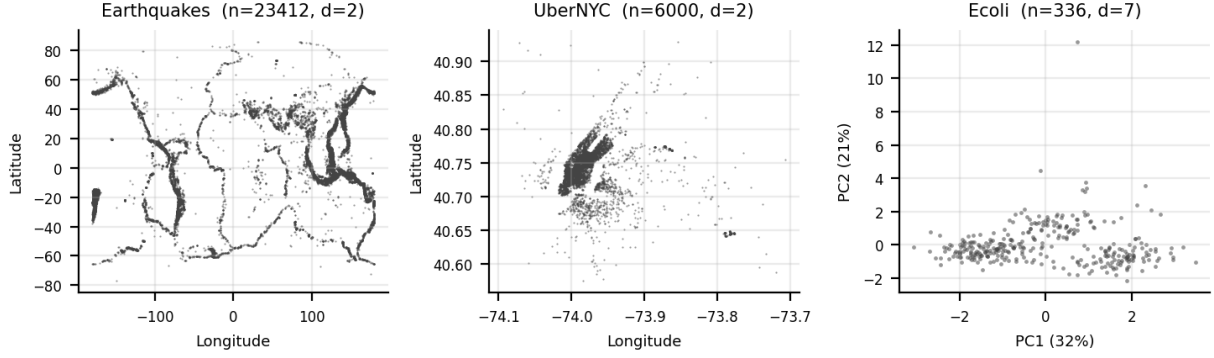


Figure 1: The three datasets. D1 and D2 are shown in geographic coordinates; D3 is projected onto its first two principal components (53% of variance). The arc-shaped, branching structure of D1 and the extreme density contrast of D2 are both visible without any clustering.

4.5 Reproducibility

All experiments use scikit-learn 1.8.0 on a single CPU core. Initialisation ablations average over 30 seeds and report mean $\pm$ standard deviation; all other runs use seed 42. Every number in this report is produced by `src/run.py` and written to `results/*.csv`; every figure by `src/figures.py`.

5 Results and Analysis

5.1 Main comparison

Table 3 gives the full comparison at the selected operating points, and Figure 2 shows the corresponding partitions.

Table 3: Main comparison at the selected operating point. Best value per dataset in **bold**. S = silhouette, DB = Davies–Bouldin, CH = Calinski–Harabasz.

Data	Method	#cl	noise	S $\uparrow$	DB $\downarrow$	CH $\uparrow$	ARI $\uparrow$	NMI $\uparrow$	time (s)
D1	K-Means ($k=3$)	3	0%	0.654	0.621	96836	—	—	0.043
D1	K-Means++ ($k=3$)	3	0%	0.650	0.632	96854	—	—	0.038
D1	DBSCAN	48	29%	0.401	0.426	134039	—	—	0.120
D1	HDBSCAN	13	32%	0.698	0.341	123748	—	—	3.331
D2	K-Means ($k=6$)	6	0%	0.487	0.634	5899	—	—	0.025
D2	K-Means++ ($k=6$)	6	0%	0.488	0.637	5900	—	—	0.026
D2	DBSCAN	3	1%	0.687	0.326	1580	—	—	0.159
D2	HDBSCAN	3	5%	0.627	0.284	3117	—	—	0.232
D3	K-Means ($k=7$)	7	0%	0.248	1.123	154	0.493	0.607	0.008
D3	K-Means++ ($k=7$)	7	0%	0.354	0.861	168	0.731	0.679	0.009
D3	DBSCAN	2	21%	0.434	0.908	220	0.489	0.488	0.003
D3	HDBSCAN	2	37%	0.483	0.789	211	0.339	0.376	0.003

Three observations follow directly.

The spatial results are structurally, not marginally, different. On D1, K-Means at its selected $k = 3$ produces three longitude wedges. These are compact and well separated in the Euclidean sense — silhouette 0.654 — and completely uninformative seismologically, because a cluster boundary drawn through the middle of the Pacific plate corresponds to no physical structure. DBSCAN at the same operating point returns 48 clusters that follow the plate boundaries, at a silhouette of 0.401. The method that scores worse on the internal index is the one that answers the

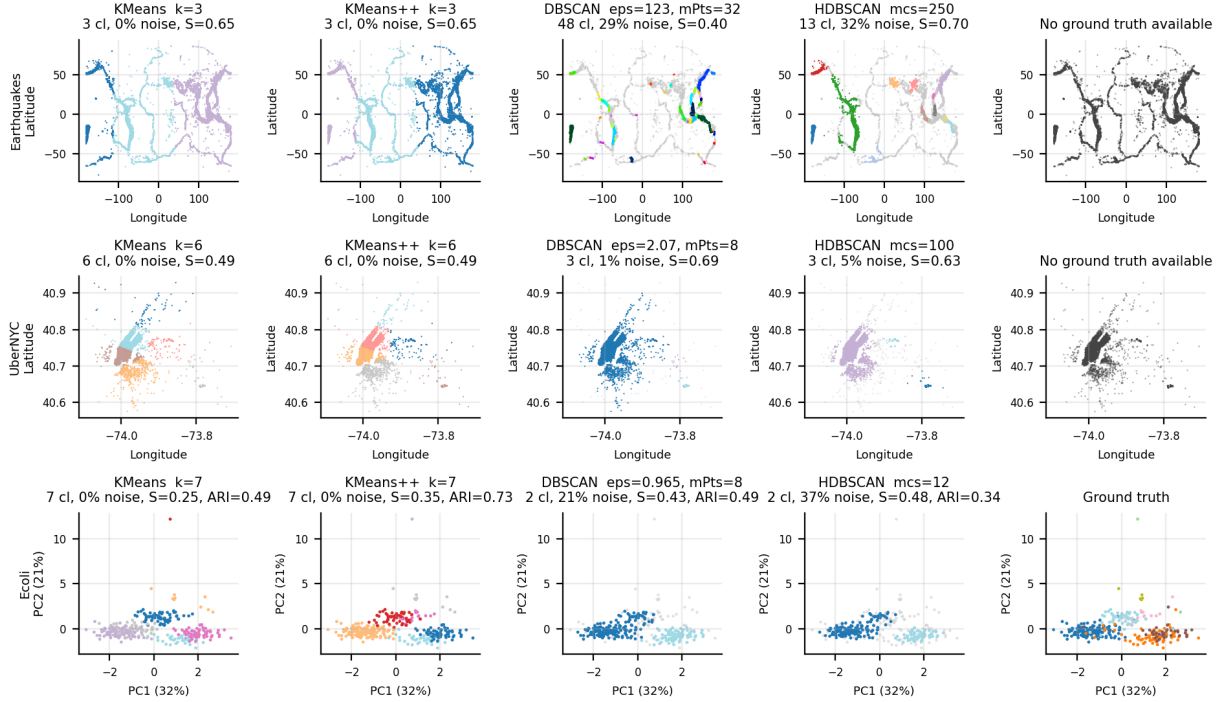


Figure 2: Qualitative comparison. Rows are datasets, columns are methods; grey points are noise. The D1 row is the clearest illustration in the study: K-Means partitions the globe into three convex wedges that slice across the Mid-Atlantic Ridge and the Pacific Ring of Fire, while DBSCAN and HDBSCAN trace the boundary arcs themselves.

question.

Internal indices do not agree with each other either. On D1, silhouette selects HDBSCAN, Calinski–Harabasz selects DBSCAN, and Davies–Bouldin selects HDBSCAN. On D2, silhouette selects DBSCAN while Calinski–Harabasz selects K-Means — by a factor of nearly four. An analyst choosing a method by internal index alone would reach a different conclusion depending on which index was consulted.

Initialisation matters more than expected on D3. K-Means and K-Means++ run at identical k with identical update rules differ by 0.24 ARI (0.493 versus 0.731). This is the single largest effect of any parameter change in the study, and it arises purely from seeding. Section 6.1 quantifies it.

5.2 Parameter sensitivity

Figure 3 shows the k sweep. On both spatial datasets the silhouette curve is monotonically decreasing from $k = 2$, so the unconstrained optimum is the smallest admissible k — the degeneracy that motivated the constrained protocol of Section 4.4. The inertia elbow, by contrast, falls at $k = 8$ (D1) and $k = 7$ (D2), which is both more plausible and more useful. On D3, where ARI is available, the ARI curve peaks at $k = 7$ against a true class count of 8; silhouette peaks at $k = 5$. Even here, where the data are relatively well behaved, silhouette underestimates the number of groups.

Figure 5 shows the DBSCAN grid. The dominant pattern is that cluster count and noise fraction are far more sensitive to ε than to minPts : across the grid, varying ε by a factor of 30 moves the cluster count over two orders of magnitude, while varying minPts from 4 to 32 changes it by less than a factor of three. The practical implication is that effort spent tuning minPts is largely wasted; ε is the parameter that matters.

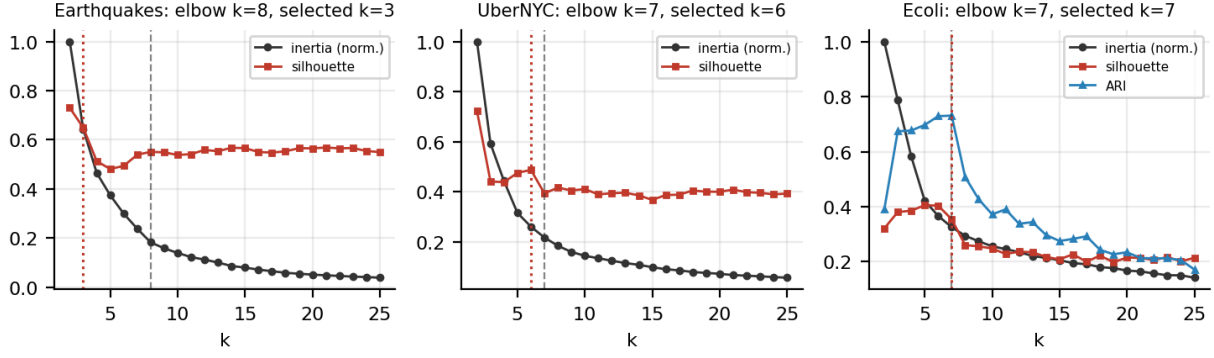


Figure 3: K-Means k sweep. Dashed line: inertia elbow. Dotted line: k selected under the constrained protocol. On D1 and D2 the silhouette declines monotonically in k , so the constraint — not the metric — determines the answer.

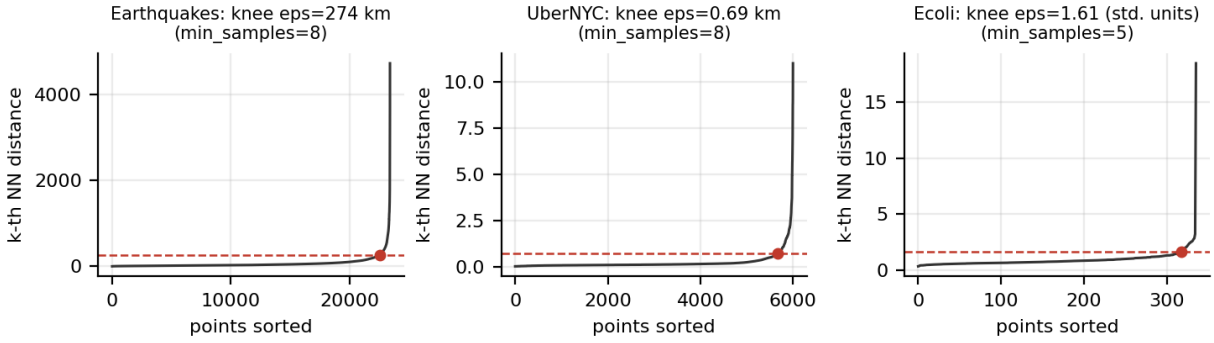


Figure 4: k -distance curves with the detected knee. The knee gives $\varepsilon = 274$ km for D1 and 0.69 km for D2 — both physically meaningful quantities, unlike k .

5.3 The operating-point problem

Figure 6 isolates the practical consequence. On D1 the k -distance knee gives $\varepsilon = 274$ km, which yields 68 clusters that trace the plate boundaries almost perfectly while discarding only 2% of events — and scores a silhouette of -0.03 , essentially the worst possible value. The silhouette-optimal setting, $\varepsilon = 123$ km, raises the score to 0.40 by fragmenting the boundaries and discarding 29% of the events. On D2 the same pattern appears in reverse: the knee at $\varepsilon = 0.69$ km separates Manhattan, Brooklyn, JFK and LaGuardia into 11 usable zones; the silhouette-optimal $\varepsilon = 2.07$ km merges them into three, collapsing all of Manhattan into a single undifferentiated blob that provides no basis for positioning vehicles.

In both cases the silhouette prefers the operationally useless answer. It does so for a coherent reason: elongated and irregular clusters have large within-cluster distances by construction, so the index penalises exactly the structure the application is trying to find.

5.4 Internal and external indices disagree systematically

D3 lets us test the above claim against ground truth, and the result is stark. Ranking the four methods by silhouette gives HDBSCAN > DBSCAN > K-Means++ > K-Means. Ranking them by ARI gives almost exactly the reverse: K-Means++ > K-Means > DBSCAN > HDBSCAN. The Pearson correlation between the two rank vectors is -0.80 (Figure 7). The best method by silhouette is the worst by ARI, and vice versa.

The mechanism is visible in the noise column of Table 3. HDBSCAN discards 37% of the proteins. Removing the sparse, ambiguous points inflates the silhouette of what remains, but those

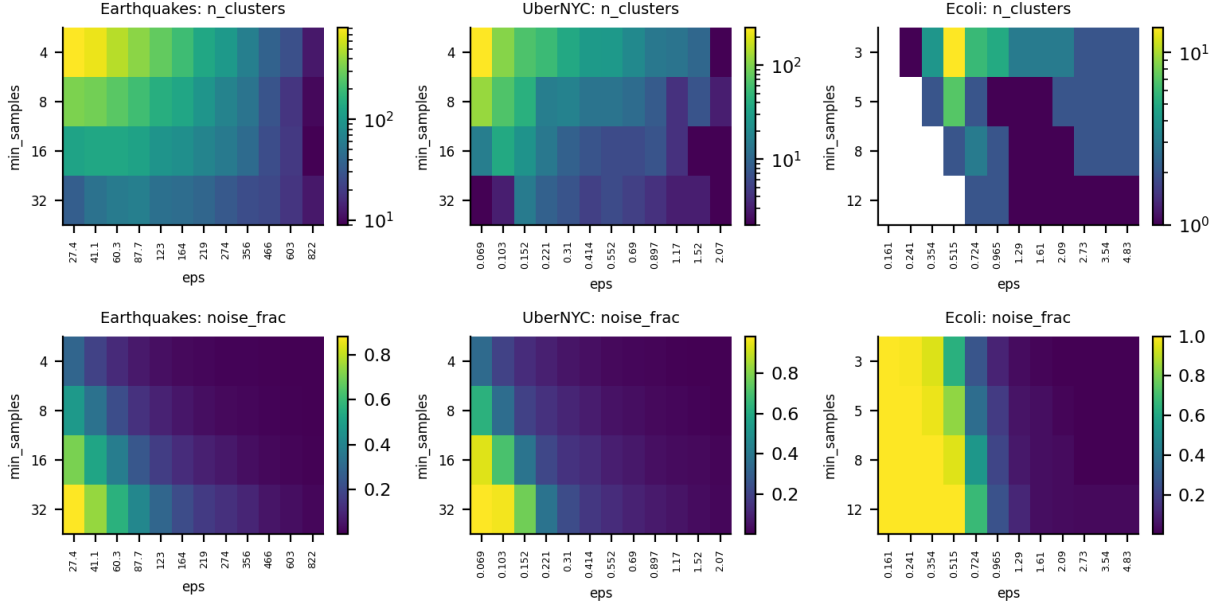


Figure 5: DBSCAN parameter grid: cluster count (top, log scale) and noise fraction (bottom). ϵ dominates; minPts is a secondary control.

discarded points include entire rare localisation classes, which is why ARI collapses to 0.339. The index rewards the method for throwing away the hardest and most biologically interesting part of the problem.

This is the study’s central result, because it transfers. On D1 and D2 we cannot compute ARI — but there is no reason to expect the distortion to be smaller there. If anything it should be larger, since the spatial clusters are far less convex than the Ecoli classes. Conclusions drawn from silhouette on unlabelled spatial data should therefore be treated as unreliable by default.

6 Ablation Studies

6.1 A: Initialisation

K-Means and K-Means++ differ only in seeding. Over 30 seeds at the selected k (Table 4, Figure 8), K-Means++ reduces the mean objective by 8.4% on D1 and by 25.2% on D3, and reduces the iteration count on all three datasets — most sharply on D3, from 17.7 to 11.0. On D2 the mean objective is statistically indistinguishable and only the iteration count improves.

The variance result is more nuanced than the standard account suggests. On D1 K-Means++ reduces the coefficient of variation from 11.0% to 9.1%, but on D3 it *increases* it from 8.4% to 12.9% — while simultaneously improving the mean by a quarter. K-Means++ is therefore not simply a variance-reduction technique here: on D3 it moves the whole distribution to a better region of the objective landscape and explores that region more widely. The practical recommendation is unchanged (use K-Means++), but the reason is the mean, not the spread.

6.2 B: Distance metric on geographic data

We compare three treatments of geographic coordinates at matched ϵ : exact haversine distance, naive Euclidean distance on raw degrees, and Euclidean distance on the projected kilometre coordinates used throughout.

On D1 all three return 48 clusters, but haversine retains more data (26.4% noise versus 29.3%) — the projection understates distances at high latitude, so marginally fewer points qualify as core

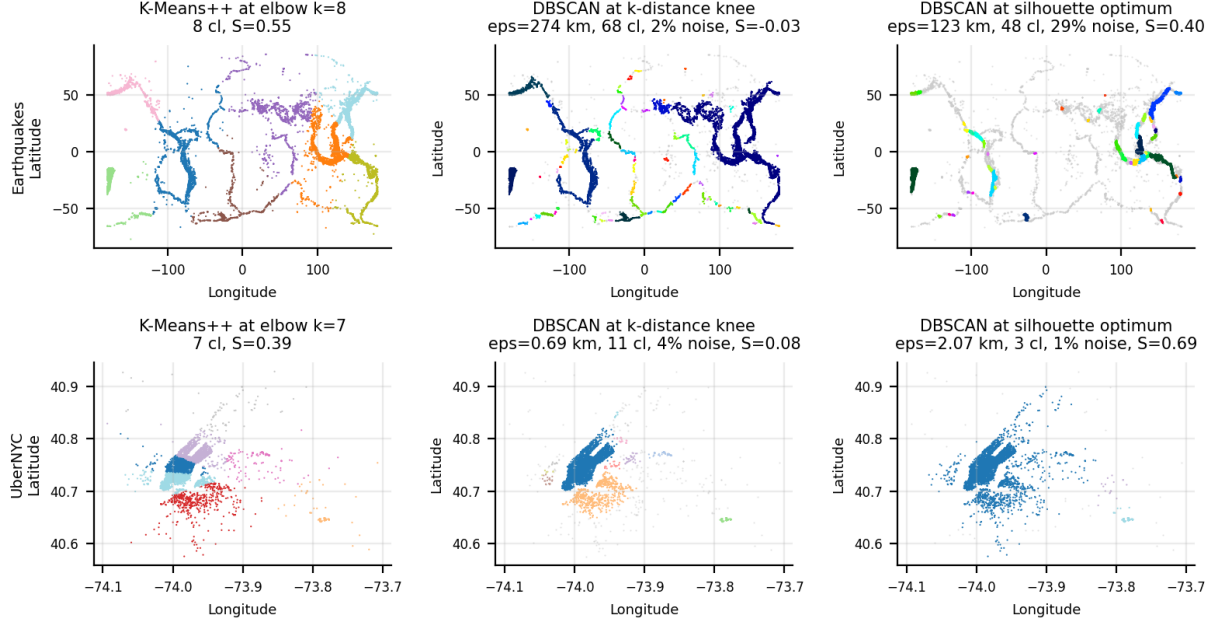


Figure 6: Operating-point selection. Left: K-Means at the inertia elbow. Centre: DBSCAN at the k-distance knee. Right: DBSCAN at the silhouette optimum. On D1 the silhouette prefers a solution that discards 29% of events over one that discards 2% and recovers the plate boundaries; on D2 it prefers three merged blobs over eleven usable demand zones.

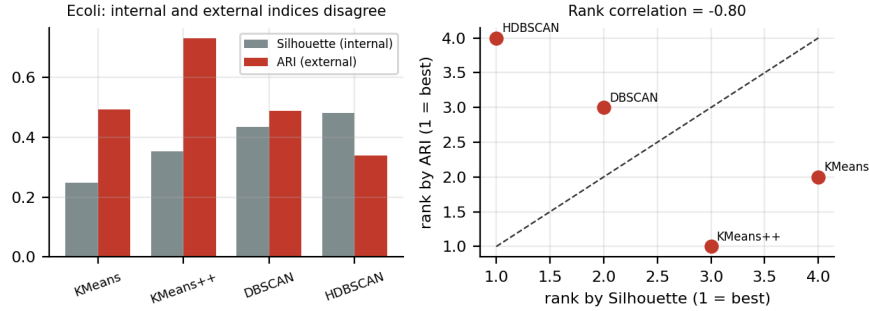


Figure 7: Ecoli: internal versus external validity. Left: raw values. Right: rank agreement; points on the dashed line would indicate agreement. The observed rank correlation is -0.80 .

points. On D2 the discrepancy is larger in relative terms: raw-degree Euclidean returns 6 clusters where haversine and projected-km return 3, because a degree of longitude at 40.7°N is only 0.76 of a degree of latitude, so a circular ε ball in degree space is an ellipse on the ground, elongated east–west. Haversine costs roughly $2\times$ the runtime of the projected version (0.40 s versus 0.13 s on D1).

The conclusion is that raw-degree Euclidean should not be used, but that a local projection is an adequate and cheaper substitute for haversine at city scale; at global scale haversine is preferable if the noise fraction matters.

6.3 C: Feature scaling

On D3, standardisation raises K-Means++ ARI from 0.497 to 0.731 , a 47% relative improvement. DBSCAN, given its own best ε in each condition, moves in the opposite direction: 0.481 to 0.455 .

The asymmetry is informative. K-Means minimises total squared distance, so a feature with larger numeric range dominates the objective and the recovered partition; equalising the ranges is therefore a direct improvement. DBSCAN only asks whether a neighbourhood contains `minPts`

Table 4: Ablation A: initialisation, 30 seeds, `n_init`=1. CV = coefficient of variation of the objective.

Data	Init	mean inertia	CV (%)	mean iters	Δ mean
D1	random	6.087×10^{11}	11.03	9.63	—
D1	k-means++	5.574×10^{11}	9.08	7.20	−8.4%
D2	random	2.825×10^4	2.40	17.83	—
D2	k-means++	2.828×10^4	2.26	11.50	+0.1%
D3	random	8.450×10^2	8.36	17.70	—
D3	k-means++	6.317×10^2	12.94	11.00	− 25.2%

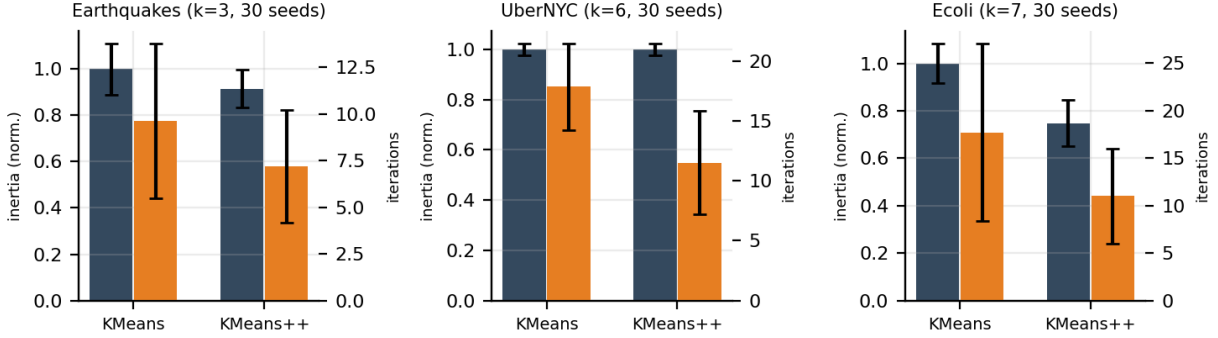


Figure 8: Ablation A. Bars show mean $\pm$ s.d. over 30 seeds: normalised objective (dark, left axis) and iterations to convergence (orange, right axis).

points, and since ε is re-tuned per condition, a global rescaling is largely absorbed by the re-tuning. Standardisation is close to mandatory for centroid methods and close to irrelevant for density methods when ε is tuned — though note this holds for *uniform* rescaling; it would not hold if only some features were rescaled.

6.4 D: Intrinsic dimensionality

We project D3 onto its first 2, 3, 4, 5 and 7 principal components and re-tune both methods in each subspace (Figure 9, Table 5). The result is a clean crossover.

Table 5: Ablation D: performance versus dimensionality on D3. “viable ε ” is the fraction of a 40-point ε grid producing ≥ 2 clusters with $< 50\%$ noise.

PCA dims	DBSCAN ARI	K-Means++ ARI	viable ε	winner
2	0.575	0.468	0.125	DBSCAN
3	0.618	0.466	0.675	DBSCAN
4	0.453	0.512	0.575	K-Means++
5	0.399	0.520	0.675	K-Means++
7	0.455	0.731	0.650	K-Means++

At two and three dimensions DBSCAN is clearly better (0.575 and 0.618 against 0.468 and 0.466). Between three and four dimensions the ranking reverses, and by seven dimensions K-Means++ leads by 0.276 ARI — a margin larger than the entire range of DBSCAN’s performance across the sweep. DBSCAN’s ARI peaks at three dimensions and then declines, while K-Means++ improves monotonically from four dimensions onward.

This is the curse of dimensionality acting on a density estimator. As d grows, pairwise distances concentrate: the ratio of the nearest to the farthest neighbour distance tends to one, so the ε -ball that separates “dense” from “sparse” becomes progressively less discriminative. K-Means is comparatively insensitive because it compares each point only to k centroids rather than

relying on local neighbourhood counts, and because the additional principal components carry class-discriminative variance that its objective can exploit.

This ablation also explains the main table. D1 and D2 are two-dimensional, where density methods are strongest; D3 is seven-dimensional, where they are weakest. The apparent contradiction between “DBSCAN wins on spatial data” and “K-Means++ wins on Ecoli” is not a contradiction at all — it is a single monotone relationship with intrinsic dimensionality, observed at two points.

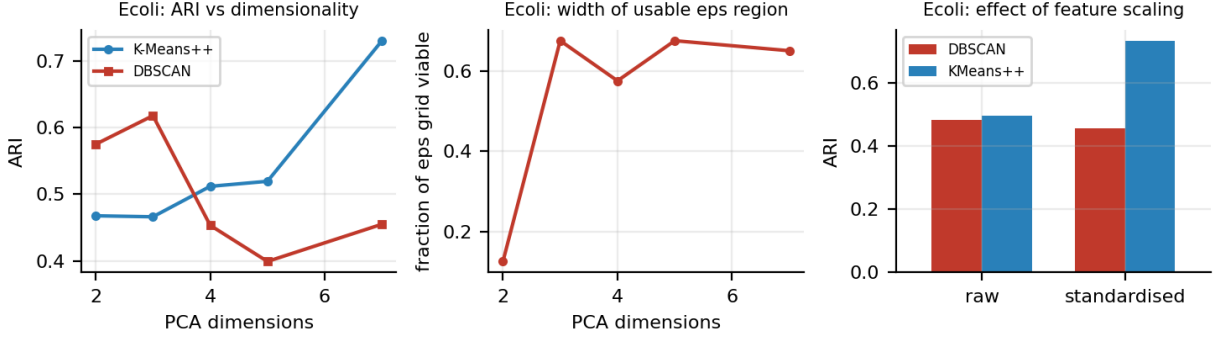


Figure 9: Ablations C and D. Left: ARI versus PCA dimensionality, showing the crossover between three and four dimensions. Centre: the usable ε region. Right: effect of standardisation, which helps K-Means++ substantially and does not help DBSCAN.

6.5 Runtime

At the selected operating points (Figure 10), K-Means is fastest on every dataset, DBSCAN costs roughly 3–6 $\times$ more, and HDBSCAN is the outlier: 3.33s on D1 against 0.04s for K-Means, an 87 $\times$ gap. At $n \leq 23,412$ none of these differences is operationally meaningful — all four methods finish in seconds — so runtime should not drive the choice at this scale. The asymptotics do differ, however: K-Means is $O(nkdi)$ and DBSCAN $O(n \log n)$ with a spatial index, so the ordering would change at much larger n with small k .

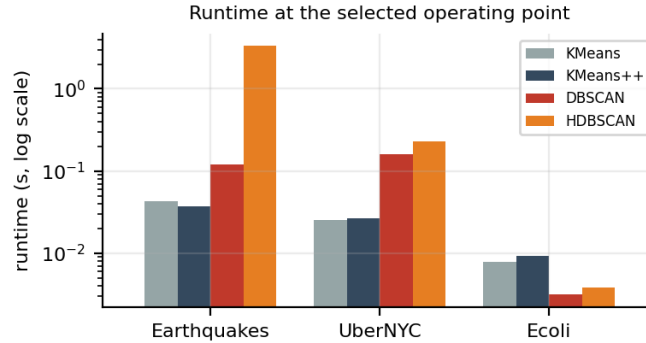


Figure 10: Runtime, log scale. HDBSCAN’s hierarchy construction dominates on D1.

7 Discussion

7.1 Successful cases

DBSCAN on D1 at $\varepsilon = 274\text{km}$ is the clearest success in the study: 68 clusters that follow the Mid-Atlantic Ridge, the East Pacific Rise and the circum-Pacific subduction zones, with 2% of events labelled noise — a figure consistent with the expected rate of intraplate seismicity. No value

of k would let K-Means produce this, because the clusters are one-dimensional curves embedded in two dimensions.

DBSCAN on D2 at the k -distance knee separates the airports from the Manhattan core, which is the operationally important distinction for a fleet operator, and does so with 4% noise.

K-Means++ on D3 is the third success, and it is a success for the opposite reason: the Ecoli classes are compact blobs in a seven-dimensional space with no meaningful density gaps between them, which is the regime K-Means was designed for.

7.2 Failure cases

K-Means on D1 is a textbook failure: three convex wedges imposed on a curvilinear structure, with high silhouette and zero scientific content. The failure is not of tuning but of representational capacity.

HDBSCAN on D3 fails by over-discarding: 37% of proteins labelled noise, with the rare localisation classes among them, giving the worst ARI in the study while simultaneously achieving the best silhouette. A method that improves its score by deleting the hard cases is dangerous precisely when the hard cases matter.

Silhouette-driven selection on D2 fails by merging: the optimum collapses Manhattan into one cluster, which is a correct statement about the geometry (the density really is continuous there) and a useless one for the application.

7.3 Key factors affecting performance

Ordered by the magnitude of their effect in our experiments: **(i) intrinsic dimensionality** — 0.28 ARI swing on D3 and a reversal of the method ranking (Section 6.4); **(ii) initialisation** — 0.24 ARI on D3 at fixed k (Section 6.1); **(iii) feature scaling** — 0.23 ARI for K-Means++ (Section 6.3); **(iv) ε** — moves cluster count over two orders of magnitude (Figure 5); **(v) cluster convexity** — decides the outcome qualitatively on D1 but cannot be read off any single number; **(vi) minPts** — a secondary control; **(vii) runtime** — irrelevant at this scale.

7.4 Practical guidance

1. **Decide by geometry, not by score.** Plot the data first. If the structure is curvilinear or branching, no amount of k -tuning will help; if it is blob-like in more than four dimensions, no amount of ε -tuning will.
2. **Never select parameters by unconstrained silhouette on non-convex data.** It reliably prefers fewer, rounder clusters and a larger discard rate. Constrain the search by what the application needs first, then optimise inside that region.
3. **Report the noise fraction alongside every internal index.** A silhouette computed after discarding 37% of the data is not comparable to one computed after discarding none.
4. **Always use k-means++ seeding.** It is free and, on D3, worth 0.24 ARI at identical k .
5. **Prefer ε to k when the units are meaningful.** “Hotspots are neighbourhoods within 700 m” is a statement a domain expert can check; “there are six hotspots” is not.
6. **Project geographic coordinates.** Raw degrees give an anisotropic distance and changed the D2 cluster count from 3 to 6.

7.5 Limitations

Our spatial evaluation rests on internal indices whose reliability this very paper questions; the D3 result gives a reason to distrust them but not a replacement. Ground truth for D1 could in

principle be constructed from published plate-boundary geometries, which we did not attempt. All datasets are at most seven-dimensional and at most 23,412 points, so the runtime comparison probes only a narrow regime. We compare four methods from two families; spectral clustering, Gaussian mixtures and OPTICS would extend the picture. Finally, the D3 oracle protocol reports each method’s best achievable ARI, which is optimistic relative to what an analyst without labels could obtain.

8 Conclusion

We compared K-Means, K-Means++, DBSCAN and HDBSCAN on two spatial hotspot-detection tasks and one labelled biological benchmark. The headline empirical result is that the standard tool for model selection in the unlabelled setting — the silhouette coefficient — ranks the four methods in nearly the reverse of their true order when ground truth is available (rank correlation -0.80), and on the spatial data prefers operating points that are demonstrably useless for the application: 29% of earthquakes discarded rather than 2%, three merged demand zones rather than eleven usable ones. The strongest single piece of evidence is Figure 6, where the setting that recovers the tectonic plate boundaries almost exactly scores a silhouette of -0.03 .

Underlying the method comparison is a single monotone factor. A dimensionality ablation on the labelled data shows DBSCAN leading below four dimensions and K-Means++ leading above it, which reconciles the apparently contradictory results across our three datasets: the two-dimensional spatial problems and the seven-dimensional biological one sit on opposite sides of one crossover.

Two directions follow. First, constructing approximate ground truth for the spatial datasets from published plate-boundary or administrative geometries would let the metric-disagreement result be tested directly where it matters most. Second, the failure modes we document — non-convexity for centroids, density variation and high dimensionality for density estimators — are exactly what spectral clustering and deep embedded clustering are designed to address, and a comparison extended in that direction is the natural next step.

Reproducibility. Code: `src/data.py`, `src/core.py`, `src/run.py`, `src/figures.py`. Results: `results/main.csv`, `sweep.csv`, `ablation.csv`, `chosen_params.json`. Data: global earthquake catalogue ($M \geq 5.5$, 1965–2016) and NYC ride-hailing pickups (April 2014), both public; Ecoli from the UCI Machine Learning Repository (Horton & Nakai, 1996).